

CUSTOMER SEGMENTATION & CHURN PATTERN ANALYTICS IN EUROPEAN BANKING

A Data Analytics Project Report

Prepared using Python, Pandas, Plotly & Streamlit

Submitted by: Ganesh Gugulothu
Date: August 2026

1. Executive Summary

Customer churn is a significant challenge in retail banking because the loss of existing customers reduces customer lifetime value, increases customer acquisition costs, and creates instability in long-term business relationships. This project, Customer Segmentation & Churn Pattern Analytics in European Banking, develops a segmentation-driven analytical framework to identify customer groups with elevated observed churn.

The analysis was conducted on a dataset of 10,000 banking customers from France, Germany, and Spain. Customer characteristics such as geography, gender, age, credit score, tenure, account balance, product usage, active membership status, and estimated salary were analyzed against customer exit behavior.

The overall observed churn rate was 20.37%, corresponding to 2,037 churned customers. Germany emerged as the highest-risk market at approximately 32.44% churn, compared with 16.15% in France and 16.67% in Spain. Age segmentation showed the 46-60 group at an especially high 51.12% churn rate, and gender analysis found higher churn among female customers (25.07%) than male customers (16.46%).

High-value customer analysis identified 2,426 premium customers (balance above 100,000), of whom 612 churned - a premium-segment churn rate of approximately 25.23%. An interactive Streamlit dashboard was built to make these findings accessible to non-technical stakeholders, with dynamic filters, KPI cards, Plotly visualizations, and downloadable data.

Overall, the analysis suggests that retention strategy should prioritize high-risk geographic and demographic segments, protect high-value relationships, and strengthen customer engagement. These findings represent descriptive associations, not proof that any single characteristic causes churn.

2. Background

Customer retention is a critical component of sustainable banking operations. While banks continuously acquire new customers, retaining existing relationships can be equally important, since customers may hold multiple products, maintain substantial balances, and contribute to long-term business value.

Traditional churn reporting often focuses primarily on a single overall churn percentage. However, one number cannot explain which customers are leaving, where churn is concentrated, or which relationships represent the greatest financial exposure. This project therefore adopts a segmentation-driven approach to churn analytics.

3. Problem Statement

The central problem addressed by this project is:

How can customer-level banking data be transformed into actionable segmentation insights that help identify customer groups with elevated observed churn and support more targeted retention strategies?

The analysis specifically investigates:

- Which geographic regions experience the highest churn?
- Which demographic groups demonstrate elevated churn?
- How does churn vary with credit profile, tenure and account balance?
- Are high-value customers experiencing significant churn?
- How does customer activity relate to churn?
- What financial exposure is associated with customers who have exited?

4. Project Goals

Primary Objectives

- Measure the overall customer churn rate.
- Identify churn patterns across customer segments.
- Compare churn across France, Germany, and Spain.
- Identify demographic groups with elevated churn.
- Develop an interactive analytical dashboard for stakeholder decision support.

Secondary Objectives

- Analyze churn among high-balance customers.
- Investigate customer engagement and tenure.
- Compare churned and retained customer profiles.
- Analyze salary and balance patterns.
- Estimate the account-balance exposure associated with churned customers.
- Support targeted retention and marketing strategies.

5. Dataset Overview

The dataset consists of 10,000 customer records. Key variables include:

Variable	Purpose
CreditScore	Customer creditworthiness
Geography	France, Germany or Spain
Gender	Customer gender
Age	Customer age
Tenure	Number of years with the bank
Balance	Account balance
NumOfProducts	Number of bank products
HasCrCard	Credit-card ownership
IsActiveMember	Customer activity indicator
EstimatedSalary	Estimated annual salary
Exited	Customer churn indicator

Churn indicator: Exited = 0 (Retained customer), Exited = 1 (Churned customer)

6. Technology Stack

Technology	Usage
Python	Core analysis
Pandas	Data manipulation and segmentation
Plotly	Interactive visualizations
Streamlit	Interactive web dashboard
Git	Version control
GitHub	Source-code repository
Streamlit Community Cloud	Dashboard deployment
VS Code	Development environment

7. Data Validation and Cleaning

Before conducting the analysis, the dataset was validated for structure, missing values, data types, and descriptive statistics using the following operations:

- `df.info()`
- `df.isnull().sum()`
- `df.describe()`

Non-analytical identifying fields (CustomerId, Surname) were removed from the analytical dataset using `df.drop(["CustomerId", "Surname"], axis=1, inplace=True)`. Removing these fields ensured that the analysis focused on behavioral, demographic, and financial characteristics rather than customer identity.

8. Feature Engineering and Customer Segmentation

A major component of the project involved transforming continuous variables into interpretable business segments.

Age Segmentation

- <30
- 30-45
- 46-60
- 60+

Credit Score Segmentation

- Low
- Medium
- High

Tenure Segmentation

- New
- Mid-term
- Long-term

Balance Segmentation

- Zero Balance
- Low Balance
- Medium Balance
- High Balance

These engineered features made it possible to compare churn behavior across meaningful customer groups rather than analyzing only individual numerical values.

9. Key Performance Indicators

Overall Customer KPIs

KPI	Value
Total Customers	10,000
Churned Customers	2,037
Overall Churn Rate	20.37%
Average Balance	~76,485.89
Average Estimated Salary	~100,090.24
Active Members	~51.51%

High-Value Customer KPIs

KPI	Value
Premium Customers	2,426
Premium Customers Churned	612
Premium Customer Churn Rate	25.23%

10. Key Analytical Findings

Finding 1 - Overall Churn Is Substantial

The overall churn rate was approximately 20.37%, meaning roughly one in every five customers in the dataset exited the bank. This result demonstrates the importance of moving beyond aggregate churn reporting toward segment-level analysis.

11. Geographic Analysis

Country	Churn Rate
Germany	32.44%
Spain	16.67%
France	16.15%

Observation: Germany shows an observed churn rate of 32.44%, roughly twice that of France (16.15%) and Spain (16.67%).

Business Interpretation: Germany represents the highest observed geographic churn risk. Applying a uniform retention strategy across all three markets would under-serve the German customer base.

Recommended Action: Investigate the German customer base more deeply, including product usage, engagement, age distribution, service experience, and high-value customer behavior, and prioritize it for targeted retention initiatives.

12. Gender Analysis

Gender	Churn Rate
Female	25.07%
Male	16.46%

Observation: Female customers show a higher observed churn rate (25.07%) than male customers (16.46%).

Business Interpretation: Gender should not be interpreted as a causal factor by itself. This finding should motivate investigation into whether other characteristics such as geography, age, products, activity, or balances differ systematically between these groups.

Recommended Action: Use gender as one input within multi-dimensional segmentation rather than as a standalone targeting variable, and investigate correlated factors before designing interventions.

13. Age Segmentation Analysis

Age Group	Churn Rate
<30	7.50%
30-45	15.74%
46-60	51.12%
60+	24.78%

Observation: The 46-60 age group records a churn rate above 50% (51.12%), substantially higher than every other age segment.

Business Interpretation: This is the single strongest observed demographic risk segment in the dataset, far exceeding younger customer segments.

Recommended Action: Investigate the needs, product usage, engagement levels, and service experience of customers aged 46-60, and design tailored retention strategies specifically for this group.

14. Credit Score Analysis

Credit Group	Churn Rate
Low	23.64%
Medium	20.29%
High	19.87%

Observation: Low-credit-score customers show somewhat higher churn (23.64%) than medium (20.29%) and high (19.87%) credit groups.

Business Interpretation: The differences between credit groups are considerably smaller than the differences observed across geography and age, so credit score should not be treated as the strongest standalone churn indicator.

Recommended Action: Include credit score as a secondary, supporting variable in segmentation models rather than a primary targeting criterion.

15. Tenure Analysis

Tenure Group	Churn Rate
New	21.15%
Mid-term	20.76%
Long-term	19.67%

Observation: Churn rates across New (21.15%), Mid-term (20.76%), and Long-term (19.67%) tenure groups are close together.

Business Interpretation: Tenure alone provides limited separation between high- and low-churn groups compared with characteristics such as geography or age.

Recommended Action: Deprioritize tenure as a standalone retention trigger; combine it with stronger discriminators such as geography, age, and balance segment.

16. Balance Analysis

Balance Segment	Churn Rate
Zero Balance	13.82%
Low Balance	34.67%

Balance Segment	Churn Rate
Medium Balance	19.88%
High Balance	25.23%

Observation: Churn is not restricted to low-value customers - Low Balance shows the highest churn (34.67%), but High Balance customers also churn at a meaningful 25.23%.

Business Interpretation: Customer value is an important dimension of retention prioritization; high-balance departures can represent significant relationship exposure even though their churn rate is not the highest.

Recommended Action: Build separate retention playbooks for the Low Balance segment (likely price/engagement sensitive) and the High Balance segment (likely relationship/service sensitive).

17. High-Value Customer Analysis

For this project, customers with balances above 100,000 were classified as premium/high-value customers.

Metric	Value
Premium Customers	2,426
Premium Customers Churned	612
Premium Customer Churn Rate	25.23%

Observation: 612 of 2,426 premium customers churned, a rate of approximately 25.23%, higher than the overall churn rate of 20.37%.

Business Interpretation: Treating all churned customers equally can hide differences in customer value; a relatively small number of high-value departures may represent substantial relationship and balance exposure.

Recommended Action: Implement proactive, white-glove retention processes specifically for premium customers, with early-warning monitoring and dedicated relationship-manager outreach.

18. Customer Engagement Analysis

The dashboard compares churn among active and inactive customers using the IsActiveMember indicator. This analysis is particularly useful because engagement is potentially more actionable than static demographic characteristics.

Where inactivity is associated with elevated churn, banks could develop early engagement interventions such as:

- Personalized communication
- Relationship-manager outreach
- Relevant product recommendations
- Loyalty programs
- Digital engagement campaigns
- Service follow-ups

19. Geography x Age Interaction Analysis

One-dimensional segmentation may hide important patterns, so geography and age were also analyzed together.

Rather than concluding "Germany has high churn," the bank can investigate which age groups within Germany contribute most strongly to observed churn. This provides a more granular foundation for targeted campaigns and is more useful than broad country-level marketing.

20. Salary vs Balance Analysis

An interactive scatter plot was developed to compare Account Balance against Estimated Salary, with customers differentiated by churn status. This analysis helps determine whether churned customers cluster within particular financial profiles.

No single financial variable should automatically be treated as the cause of churn. Customer behavior is better examined as the interaction of financial, demographic, and engagement characteristics.

21. Churned vs Retained Customer Profiles

The dashboard directly compares churned and retained customers using average values for:

- Age
- Credit score
- Balance
- Estimated salary
- Tenure
- Number of products

This provides management with a concise profile-level comparison and supports the identification of variables that may deserve deeper analysis.

22. Financial Exposure Analysis

The dashboard calculates the total account balance associated with churned customers. Account balance is not equivalent to bank revenue, so this metric should be interpreted as customer balance exposure associated with churn rather than actual revenue lost. This distinction makes the analysis more technically credible.

23. Interactive Streamlit Dashboard

The Streamlit dashboard transforms the analytical workflow into an interactive decision-support application. Users can dynamically filter customers by:

- Geography
- Gender
- Age group
- Credit-score group
- Balance group
- Tenure group

All KPIs and charts automatically update according to the selected customer segment.

Dashboard Modules

KPI Overview: Total customers, churned customers, churn rate, average balance, average salary, and active-member percentage.

Overall Churn Analysis: Retained vs churned customer distribution.

Geographic Analysis: Country-level churn comparison.

Demographic Analysis: Gender and age-group churn.

Financial Profile Analysis: Credit-score and balance-segment churn.

High-Value Customer Explorer: Premium customer population, churn, and geographic distribution.

Engagement Analysis: Active vs inactive customer churn.

Geography x Age Analysis: Multi-dimensional customer segmentation.

Salary vs Balance Analysis: Financial-profile visualization by churn status.

Customer Profile Comparison: Churned vs retained characteristics.

Financial Exposure: Balances associated with churned customers.

Data Explorer: Filtered customer-level data with CSV download capability.

24. Major Business Insights

Key Management Insights

- Overall churn is 20.37%, meaning approximately one in five customers exited.
- Germany is the highest-risk geographic market, with approximately 32.44% observed churn.
- Customers aged 46-60 form the highest-risk age segment, with approximately 51.12% churn.
- Female customers show higher observed churn than male customers in this dataset.
- High-value customers are not immune to churn - 612 premium customers exited.
- Tenure shows comparatively smaller differences between customer groups.
- Retention prioritization should consider customer value and engagement in addition to raw churn counts.

25. Recommendations

1. Prioritize the German Market

Germany should receive deeper investigation and targeted retention initiatives because it has the highest observed churn rate.

2. Focus on the 46-60 Segment

Develop targeted engagement strategies for customers aged 46-60, particularly within high-risk geographic segments.

3. Protect High-Value Customers

Introduce proactive retention processes for premium customers, including relationship-manager outreach, personalized service, loyalty benefits, and early-warning monitoring.

4. Build Engagement-Based Interventions

Inactive customers should be monitored and, where appropriate, targeted through relevant communication and engagement campaigns.

5. Use Multi-Dimensional Segmentation

Avoid targeting customers using only one variable. Rather than "target German customers," use compound segments such as Germany + 46-60 + Inactive + High Balance for more focused retention strategies.

6. Develop an Early-Warning System

Extend the descriptive dashboard into a predictive churn model (Logistic Regression, Decision Tree, Random Forest, or Gradient Boosting) that estimates each customer's churn probability so retention teams can intervene before departure.

7. Collect Additional Customer Experience Data

Future datasets should include transaction frequency, mobile/online banking usage, complaints, service interactions, satisfaction scores, product profitability, transaction value, branch visits, and marketing response history.

26. Limitations

- The dataset represents customer characteristics but contains limited behavioral and customer-experience information.
- Account balance cannot be directly interpreted as bank revenue.
- The analysis identifies associations rather than causal relationships - for example, higher churn among a particular age or gender segment does not establish that age or gender caused the churn.
- The premium-customer threshold of 100,000 is an analytical definition used for this project and may differ from a bank's actual customer-value classification.
- The current system primarily provides descriptive and diagnostic analytics rather than individual-level predictive churn probabilities.

27. Future Scope

Predictive Churn Modeling

Train machine-learning models to estimate individual customer churn probabilities.

Customer Risk Score

Create a score such as Low Risk / Medium Risk / High Risk based on multiple customer characteristics.

Explainable AI

Use feature-importance or SHAP-based explanations to understand why a predictive model considers a customer high-risk.

Customer Lifetime Value

Combine churn probability with estimated customer value using a prioritization concept such as: Retention Priority = Churn Risk x Customer Value. This would help distinguish between customers who are merely likely to leave and customers whose departure may have larger business consequences.

Retention Recommendation Engine

A future system could recommend actions such as relationship-manager contact, loyalty offers, product review, personalized communication, or service follow-up according to each customer's risk profile.

28. Conclusion

This project demonstrates how customer-level banking data can be transformed into actionable churn intelligence through data validation, feature engineering, segmentation, exploratory analysis, interactive visualization, and business interpretation.

The analysis identified an overall churn rate of 20.37% and highlighted substantial variation across customer segments. Germany demonstrated the highest observed geographic churn, while customers aged 46-60 represented the highest-risk age segment. The analysis also showed that churn affects high-value customers, with 612 premium customers exiting in the analyzed dataset.

Beyond identifying churn patterns, the project developed an interactive Streamlit dashboard that enables stakeholders to dynamically explore customer segments across geography, demographics, tenure, credit profile, account balance, and engagement - transforming static analysis into a practical decision-support tool.

The findings suggest that effective churn management should move beyond generic retention campaigns toward targeted, value-aware, and engagement-focused strategies. Future extensions could integrate predictive machine learning, customer lifetime value, and explainable risk scoring to create a proactive churn early-warning system.

Overall, the project demonstrates an end-to-end data analytics workflow connecting raw customer data with interpretable business insights and actionable retention recommendations.

29. References & Links

[GitHub Repository](#)

[Live Dashboard](#)